

# Connected Okra Farm

Features & Team Split (v4: Outdoor + Optional Indoor) · Track 2: Hack Connected Everywhere

🌱 **Setting: Outdoor farms are the main option:** low-cost sensors, solar power and sunlight. **Indoor farms are an optional add-on** in empty rural buildings such as closed schools (廃校), using smart LED lighting to grow okra in the off-season. All farms share one LoRa network, one dashboard and one marketplace.

🇯🇵 **Japanese problems tackled:** 🧑 labour shortage · 📶 weak rural connectivity · 🗑 food loss · 🏠 empty rural buildings

🔗 **How the modules connect:** 🌱 Monitor → 💡 Control → ✂ Harvest → 🛡 Protect health → 📈 Sell before it spoils.

Legend: 🌞 **Outdoor** 🏠 **Indoor** 🌱 **Both** · **MVP** build for the hackathon · **Roadmap** future work · **Bonus** if time allows

## 🌱 Two tiers, one platform

|                     | 🌞 Outdoor (Basic)                                 | 🏠 Indoor (Premium add-on)                                            |
|---------------------|---------------------------------------------------|----------------------------------------------------------------------|
| <b>Who it's for</b> | Most farmers, especially small and remote farms   | Farmers or towns wanting extra, off-season supply                    |
| <b>Cost</b>         | 💰 Low: sensors, gateway, camera                   | 💰💰💰 Higher: lights, fans, pump, building setup                       |
| <b>Light</b>        | 🌞 Sunlight                                        | 💡 Smart LED lighting                                                 |
| <b>Power</b>        | 🔋 Battery + solar                                 | ⚡ Mains electricity                                                  |
| <b>Main value</b>   | Save labour, pick at the right time, sell surplus | Grow okra in the off-season when supply is low and prices are higher |

📅 **17 Outdoor farms supply in summer** 🌞 → **indoor farms continue in the off-season** ❄ → **buyers get okra all year**

## 1 🌱 Farm Monitor

*Senses soil and environment conditions and sends data over LoRa.*

| Feature                           | Description                                                                                                        | Example 🍅                               | Where         | Scope      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------|------------|
| 📶 <b>Farm sensors</b>             | ESP32 nodes measure soil moisture, temperature, humidity and light level.                                          | Field A: soil 42%, 29°C, 65% RH         | 🌱 <b>Both</b> | <b>MVP</b> |
| 🔧 <b>Soil nutrient monitoring</b> | Sensors track soil pH and nutrient level (EC) and warn when fertiliser is needed.                                  | "Field B nutrients low, add fertiliser" | 🌱 <b>Both</b> | <b>MVP</b> |
| 🔗 <b>LoRa farm network</b>        | Outdoor fields and indoor farms across a rural area send data to one shared gateway, even with weak mobile signal. | 4 fields + 1 indoor farm → 1 gateway    | 🌱 <b>Both</b> | <b>MVP</b> |
| 💾 <b>Offline data backup</b>      | The gateway saves readings when internet is down and uploads them later.                                           | Signal lost 3 hours → nothing lost      | 🌱 <b>Both</b> | <b>MVP</b> |

|                                                                                                              |                                                                                     |                                               |                                                                                                    |                       |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------|
|  <b>Solar-powered nodes</b> | Outdoor sensor nodes run on battery and a small solar panel, with no wiring needed. | Node runs for months without a battery change |  <b>Outdoor</b> | <b>MVP</b>            |
|  <b>Climate monitoring</b>  | Graphs of soil moisture, temperature, humidity and light throughout the day.        | Temperature peaks at 2 PM in Field A          |  <b>Both</b>    | <b>MVP</b>            |
|  <b>Weather forecast</b>    | Combines weather forecast data with sensor readings to plan watering and harvest.   | "Rain tomorrow, skip watering today"          |  <b>Outdoor</b> | <b>MVP</b>            |
|  <b>Disaster alert</b>      | Warns the farmer and neighbours of flood or landslide risk during heavy rain.       | "Hillside field saturated, stay away today"   |  <b>Outdoor</b> | <b>MVP</b><br>(basic) |
|  <b>Facility alert</b>      | Warns of power cuts, pump failure, light failure or a sensor going offline.         | "Indoor Farm 1: pump not responding"          |  <b>Indoor</b>  | <b>MVP</b>            |

## 2 Smart Control

Water for every farm; light and climate for indoor farms, automatically.

| Feature                                                                                                     | Description                                                                                        | Example  | Where                                                                                               | Scope      |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------|
|  <b>Smart irrigation</b> | Outdoor: alerts when soil is too dry. Indoor: a pump waters automatically.                         | Soil below 30% → pump runs 20 seconds                                                       |  <b>Both</b>   | <b>MVP</b> |
|  <b>Smart lighting</b>   | LED lights turn on and off automatically on a daily growth schedule.                               | 14 hours on, 10 hours off                                                                   |  <b>Indoor</b> | <b>MVP</b> |
|  <b>Light check</b>      | A light sensor confirms the plants get enough light and warns if an LED fails.                     | "Row 2 light 40% lower, check LED"                                                          |  <b>Indoor</b> | <b>MVP</b> |
|  <b>Smart energy</b>     | Lights run more during cheaper night-time electricity hours; energy use is shown on the dashboard. | "Saved ¥2,400 this month"                                                                   |  <b>Indoor</b> | <b>MVP</b> |
|  <b>Climate control</b>  | Fans turn on automatically when too hot or humid; gentle airflow also helps pollination.           | Humidity 85% → fan on for 20 min                                                            |  <b>Indoor</b> | <b>MVP</b> |
|  <b>Remote control</b>   | The farmer switches lights, pump or fans on and off from the dashboard or LINE.                    | Tap "Water now" for Indoor Farm 1                                                           |  <b>Both</b>   | <b>MVP</b> |
|  <b>Camera sync</b>      | Lights switch to full brightness when the camera takes a photo, for consistent AI detection.       | Every snapshot has even light                                                               |  <b>Indoor</b> | <b>MVP</b> |

 **Tech note:** Watering, light and fan rules run **locally on the ESP32**, so farms keep working even when the network is down. Remote commands are sent down through the LoRa gateway.

### 3 Smart Harvest & Quality

Camera + AI tells the farmer what to pick today and its grade.

| Feature                                                                                                         | Description                                                                                                        | Example 🍷                             | Where                                                                                         | Scope                       |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------|
|  <b>Farm camera</b>            | A fixed camera takes crop photos. Outdoor cameras run on solar and shoot at fixed times so sunlight looks similar. | Outdoor: photo every day at 10 AM     |  Both      | MVP                         |
|  <b>Edge AI</b>                | AI checks photos on the farm and sends only the results over LoRa.                                                 | Sends "8 flowers, 12 ready pods"      |  Both      | MVP                         |
|  <b>Flower countdown</b>       | Detects new flowers and predicts when each pod will be ready to pick.                                              | Flower opens Monday → "ready Thu–Fri" |  Both      | MVP                         |
|  <b>Pod maturity detection</b> | AI labels pods as ready, too small or overgrown.                                                                   | "12 ready, 5 too small, 2 overgrown"  |  Both      | MVP                         |
|  <b>Phone photo check</b>      | The farmer takes a photo of a pod to check its size and readiness.                                                 | "7.5 cm, ready ✅"                     |  Both      | MVP                         |
|  <b>Daily harvest map</b>    | Shows which fields, farms and rows to harvest today, ranked by urgency.                                            | "Pick Field A Row 2 and Indoor Row 5" |  Both    | MVP                         |
|  <b>Wildlife alert</b>       | The camera warns the farmer when animals enter the field.                                                          | "Animal detected at 5:40 AM"          |  Outdoor | Bonus                       |
|  <b>Quality grading</b>      | AI sorts pods into grades by size, shape and colour.                                                               | Grade A: straight, bright green       |  Both    | Roadmap                     |
|  <b>Freshness tracking</b>   | Records harvest time and flags batches that need to be delivered first.                                            | "Batch B picked 30 hours ago"         |  Both    | Roadmap<br>timestamp in MVP |

### 4 Crop Health

Warns of disease before it is visible, then confirms with the camera.

| Feature                                                                                                            | Description                                                                        | Example 🍷                            | Where                                                                                      | Scope          |
|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------|----------------|
|  <b>Disease risk prediction</b> | Warns of disease risk based on humidity, soil moisture, temperature and rain data. | 4 humid days → "High fungal risk"    |  Both | MVP            |
|  <b>Visual confirmation</b>     | The camera detects visible symptoms such as spots or yellow leaves.                | "Yellow leaves on 5 plants in Row 4" |  Both | MVP<br>(basic) |
|  <b>AI agriculture advisor</b>  | Suggests care actions based on sensor data, weather and crop condition.            | "Reduce watering, rain expected"     |  Both | MVP            |

|                                                                                                                 |                                                                   |                         |                                                                                          |             |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------|-------------|
|  <b>Yield impact</b>           | Updates the harvest forecast when plants are affected by disease. | "Friday: 45 kg → 38 kg" |  Both | MVP (basic) |
|  <b>Disease classification</b> | AI identifies the specific disease from photos.                   | "Likely powdery mildew" |  Both | Roadmap     |

## 5 Market Intelligence

*Sells the harvest before it spoils, all year round.*

| Feature                                                                                                                    | Description                                                                                        | Example 🍌                              | Where                                                                                      | Scope       |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------|-------------|
|  <b>Yield forecast</b>                    | Predicts how much okra outdoor and indoor farms will harvest in the coming days.                   | "Expected next week: ~120 kg"          |  Both   | MVP         |
|  <b>Year-round supply plan</b>            | Combines outdoor summer harvest and indoor off-season harvest into one supply calendar for buyers. | "Summer: outdoor · Off-season: indoor" |  Both   | MVP (basic) |
|  <b>Surplus alert</b>                     | Warns the farmer early when harvest will be more than orders.                                      | "~20 kg extra on Friday"               |  Both   | MVP         |
|  <b>Marketplace</b>                     | Farmers list upcoming harvest and buyers reserve it in advance.                                    | Restaurant reserves 10 kg for Saturday |  Both | MVP         |
|  <b>Surplus &amp; overgrown listing</b> | Extra or overgrown okra is sold cheaper to restaurants and processors.                             | 15 kg overgrown → okra powder maker    |  Both | MVP         |
|  <b>Demand forecasting</b>              | Predicts buyer demand based on past sales and seasons.                                             | "Demand rises 30% in August"           |  Both | Roadmap     |

## Shared

|                                                                                                               |                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|  <b>Web dashboard</b>      | One responsive website for farmers and buyers, on phone or laptop, showing outdoor and indoor farms together. |
|  <b>LINE notifications</b> | Alerts, harvest maps and remote controls are sent to farmers through LINE.                                    |
|  <b>Backend</b>            | Supabase database, auth and API that all modules share.                                                       |

## Business model

| Plan                                                                                                       | What's included                                                      | Who pays        |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------|
|  <b>Basic (Outdoor)</b> | Sensor + gateway rental, monitoring, alerts, harvest AI, crop health | Farmer, monthly |

|                                                                                                                  |                                                                  |                         |
|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------|
|  <b>Premium (Indoor add-on)</b> | Smart lighting, irrigation and climate control, energy dashboard | Farmer or town, monthly |
|  <b>Marketplace fee</b>         | Commission on matched and surplus orders                         | Buyer, per order        |

## ⚠ Challenges to prepare for (answer before judges ask)

| Challenge                                                                                                                   | Our answer                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|  <b>Indoor is expensive</b>                | Indoor is optional; outdoor is the low-cost main tier. Indoor targets off-season supply, when higher prices help cover the cost. |
|  <b>Okra grows tall (indoor)</b>           | Use compact/dwarf varieties and pruning.                                                                                         |
|  <b>Changing sunlight (outdoor camera)</b> | Outdoor snapshots are taken at fixed times; the phone photo check is a backup.                                                   |
|  <b>Radio law (技適)</b>                     | Use only 920 MHz LoRa modules certified for Japan.                                                                               |
|  <b>Existing okra AI in Japan</b>          | An Okinawa system grades okra <b>after</b> harvest; ours predicts readiness <b>before</b> picking and links it to selling.       |

## 👥 Team Split (3 people)

|                         |  <b>Person A: IoT &amp; Hardware</b>                                                                                                                                                                                                           |  <b>Person B: AI &amp; Vision</b>                                                                                                                                                                                                                      |  <b>Person C: Web &amp; Market</b>                                                                                                                                                                                                                                                                        |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main modules</b>     | <ol style="list-style-type: none"> <li>1 Farm Monitor</li> <li>2 Smart Control</li> </ol>                                                                                                                                                                                                                                       | <ol style="list-style-type: none"> <li>3 Smart Harvest &amp; Quality</li> </ol>                                                                                                                                                                                                                                                         | <ol style="list-style-type: none"> <li>5 Market Intelligence</li> </ol>                                                                                                                                                                                                                                                                                                                      |
| <b>Crop Health part</b> |  Disease risk rules                                                                                                                                                                                                                          |  Visual confirmation                                                                                                                                                                                                                                 |  Yield impact +  AI advisor                                                                                                                                                                                        |
| <b>Tasks</b>            | <ul style="list-style-type: none"> <li>• Wire ESP32 + soil, pH/EC, temp, humidity, light sensors</li> <li>• Solar + battery for outdoor nodes</li> <li>• LoRa nodes + shared gateway</li> <li>• Offline buffer + batch upload</li> <li>• Indoor: relay + LEDs, pump, fans</li> <li>• Disaster + facility alert rules</li> </ul> | <ul style="list-style-type: none"> <li>• Build okra dataset (photos + Roboflow)</li> <li>• Train YOLO: flowers, pods, leaf symptoms</li> <li>• Mount camera + Raspberry Pi</li> <li>• Run YOLO on the Pi (edge AI)</li> <li>• Outdoor fixed-time capture + indoor light sync</li> <li>• Flower countdown + harvest map logic</li> </ul> | <ul style="list-style-type: none"> <li>• Supabase setup</li> <li>• Responsive dashboard with outdoor/indoor views</li> <li>• Remote control + energy screens</li> <li>• Weather API integration</li> <li>• Yield forecast, supply calendar, surplus alert</li> <li>• Marketplace + surplus listing</li> <li>• AI advisor + LINE bot</li> <li>• Integrate everything into one demo</li> </ul> |
| <b>Skills</b>           | Arduino/C++, electronics, LoRa                                                                                                                                                                                                                                                                                                  | Python, YOLO, OpenCV, Raspberry Pi                                                                                                                                                                                                                                                                                                      | React/Next.js, SQL, APIs, LLM API                                                                                                                                                                                                                                                                                                                                                            |

## 🤝 Everyone together

|                                                                                                       |                                                                 |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
|  <b>Pitch</b>      | Pitch deck and demo video                                       |
|  <b>Validation</b> | Interview okra farmers (outdoor and indoor interest) and buyers |
|  <b>Testing</b>    | Final end-to-end testing of the full flow                       |

## 🔗 Handoff points (agree on day 1!)

| From → To | What's passed                                                                                               | Why it matters                          |
|-----------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| A → C     | Sensor data format, e.g. farm_id, type (outdoor/indoor), soil_moisture, ph, ec, temp, humidity, light, time | Dashboard, advisor and forecast need it |
| C → A     | Command format, e.g. farm_id, device (light/pump/fan), on/off, level                                        | Remote control from dashboard and LINE  |
| A → B     | Temperature data + light trigger signal                                                                     | Flower countdown and camera sync        |
| B → A     | Result message, e.g. row, flowers, ready, overdue                                                           | Must be small enough for LoRa           |
| B → C     | Detection results + harvest map data                                                                        | Feeds yield forecast and dashboard      |

⚠️ **Tip:** A and B should finish their data formats early and send test data to C before the hardware and model are ready.

## 🎬 Tabletop demo moments

| Action                           | What judges see                                   | Tier       |
|----------------------------------|---------------------------------------------------|------------|
| 💧 Let the soil dry               | Dry-soil alert appears on dashboard and LINE      | ☀️ Outdoor |
| 🌊 Pour water on the soil sensor  | Saturation → disaster alert                       | ☀️ Outdoor |
| 💡 Cover the light sensor         | LEDs turn on automatically                        | 🏠 Indoor   |
| 🌀 Breathe on the humidity sensor | Fan turns on + disease risk rises                 | 🏠 Indoor   |
| 📷 Show okra to the camera        | "12 ready pods" appears on the harvest map        | 👤 Both     |
| 📈 Forecast updates               | Surplus alert → buyer reserves on the marketplace | 👤 Both     |

🎯 Every okra connected. Every harvest sold. 🥒